ITDS231
Lecture 2
Introduction to Physical Layer
Part 1.2

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Some slides from
- ITCS323/2021s1
- NETACAD Network Essentials/ITN7.02

มหาวิทยาลัยมหิดล
Mahidol University

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 10 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

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Outline

Part 1.1

- Review: TCP/IP Protocol Suite, Communication Model, Data
- Data and Signals
- Introduction to Physical Layer
- Bandwidth and Throughput
- Transmission Media
- Network Representations and Topologies
- Guided Transmission Media

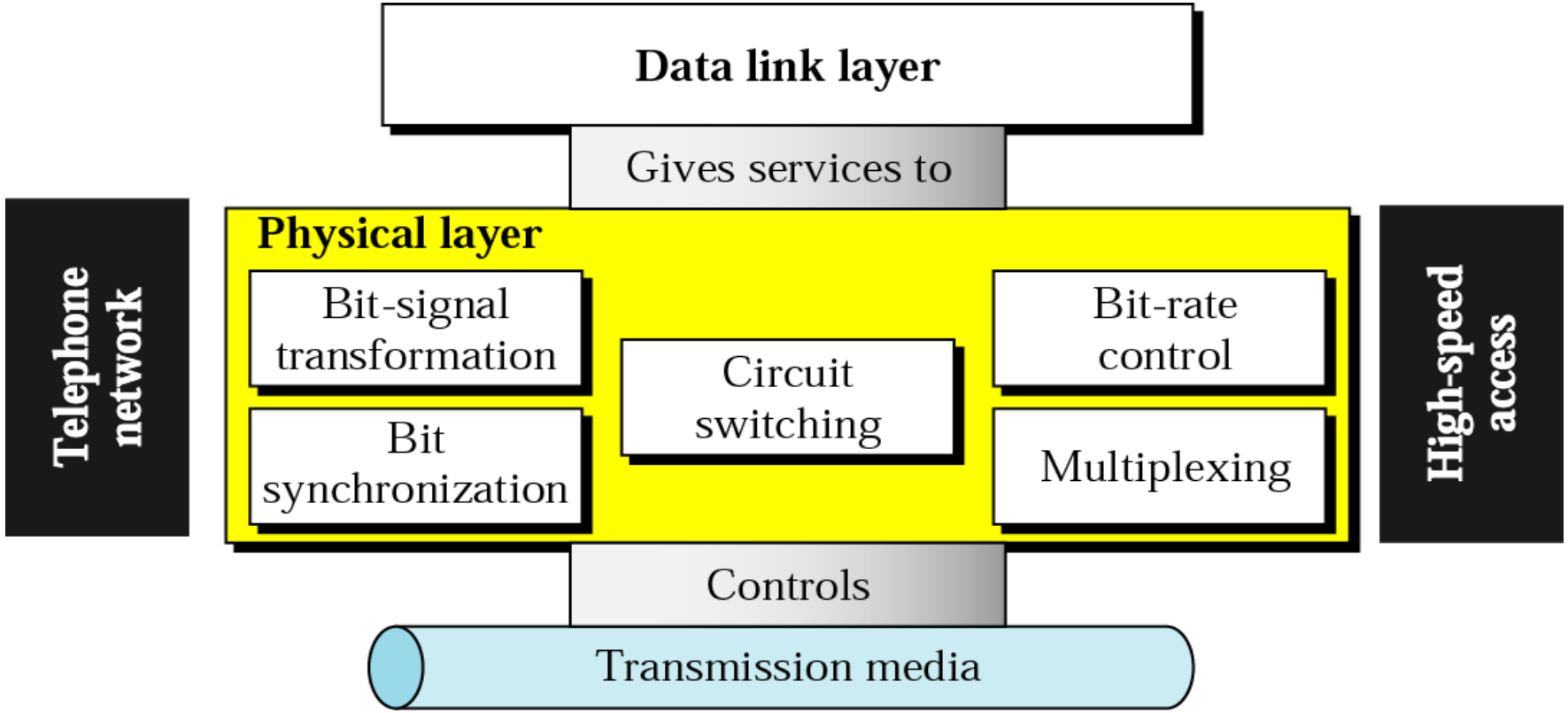
Part 1.2

- Wireless Transmission Media
- Transmission Impairment
- From waveforms to bits
- Data Rate vs Signal Rate
- Data Rate Limit

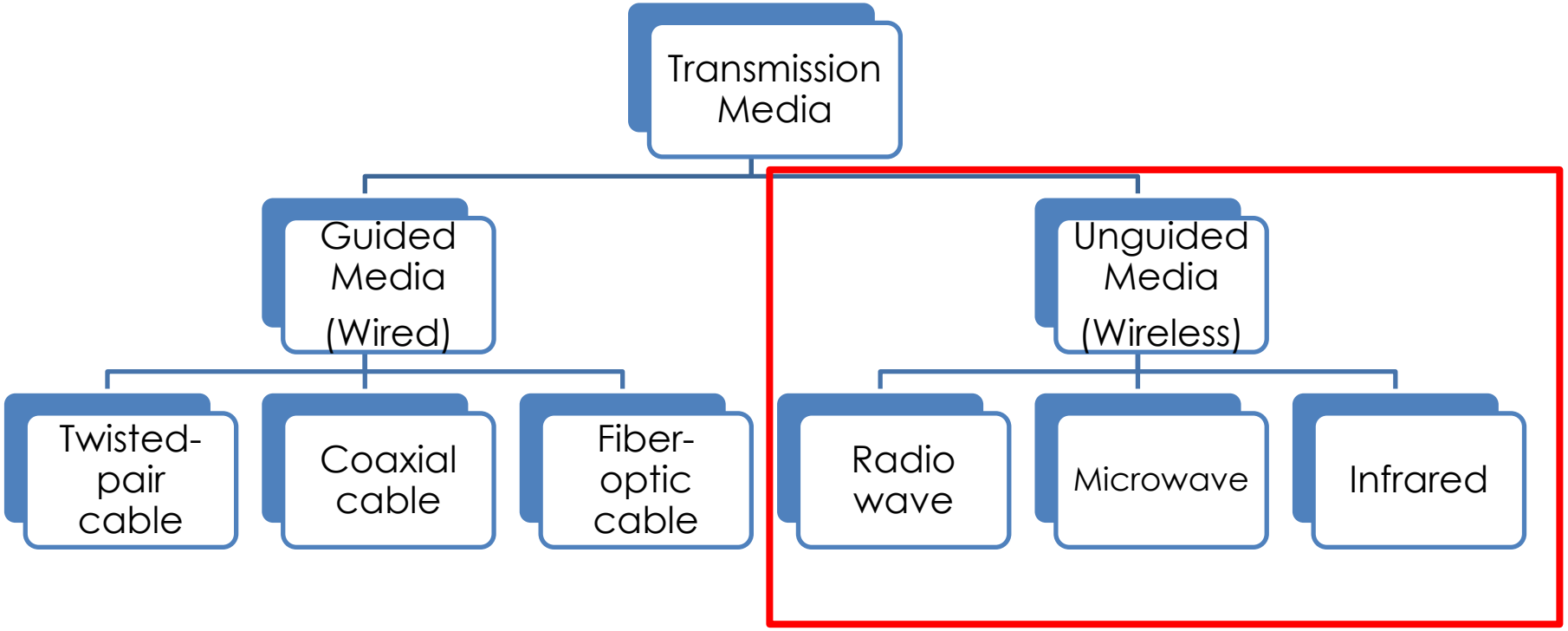
Part 2

- Digital/Analog Transmission
- Bandwidth Utilization (Multiplexing)
- Introduction to Switching
- The Public Switched Telephone Network (PSTN)
- Cellular Networks
- Cable Networks
- Communication Satellites
- Comparing Different Access Networks
- Policy of the Physical Layer
- Network Topology

Review: Functions of Physical Layer



Guide/Unguided Transmission Media

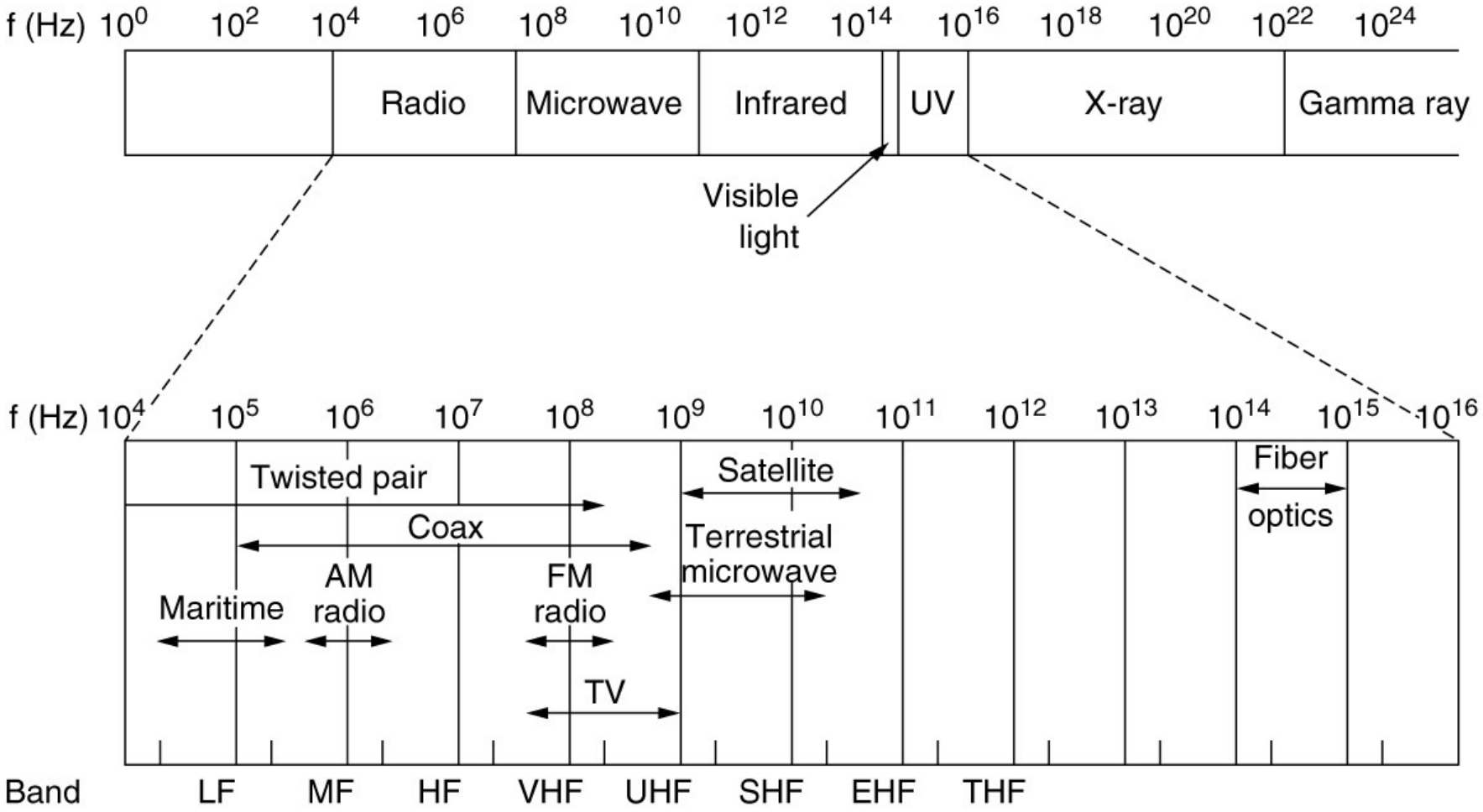


Wireless (Unguided) Transmission Media

- The electromagnetic spectrum
  - Modulate wave amplitude, frequency, or phase
- Frequency hopping spread spectrum
  - Transmitter hops from frequency to frequency hundreds of times per second
- Direct sequence spread spectrum
  - Code sequence spreads data signal over wider frequency band
- Ultra-wideband communication
  - Communication sends a series of low-energy rapid pulses, varying their carrier frequencies to communicate information

Wireless Transmission Media

The Electromagnetic Spectrum and Its uses for Communication

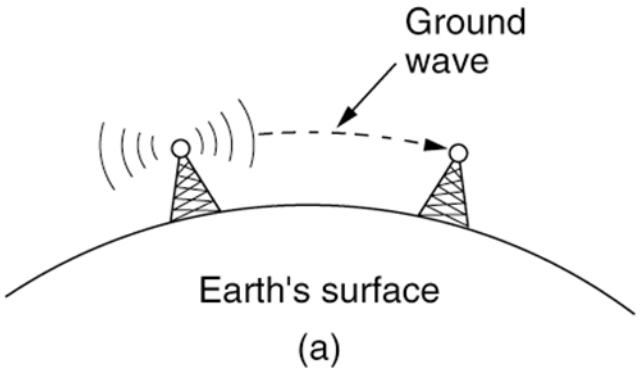


Wireless Transmission

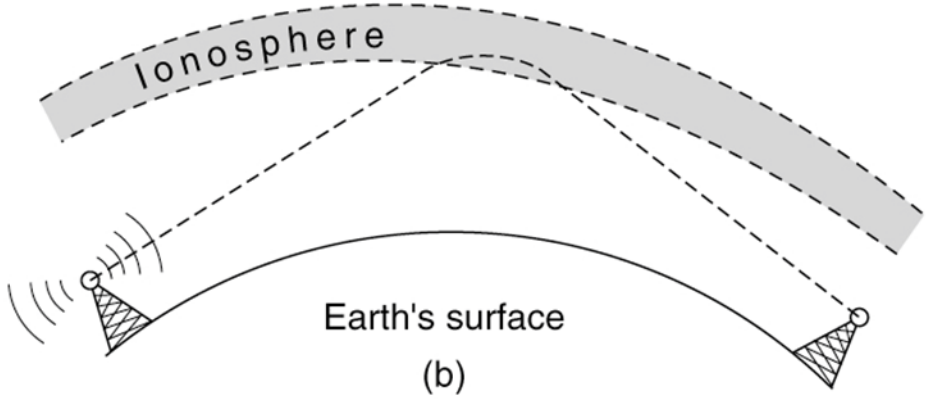
Using the Spectrum for Transmission

- Radio transmission
  - Omnidirectional waves, easy to generate, travel long distances, penetrate buildings
- Microwave transmission
  - Directional waves requiring repeaters, do not penetrate buildings
- Infrared transmission
  - Unguided waves used for short-range communication, relatively directional, cheap, easy to build, do not penetrate solid walls
- Light transmission
  - Unguided optical communication

Radio Transmission



VLF, LF, MF band

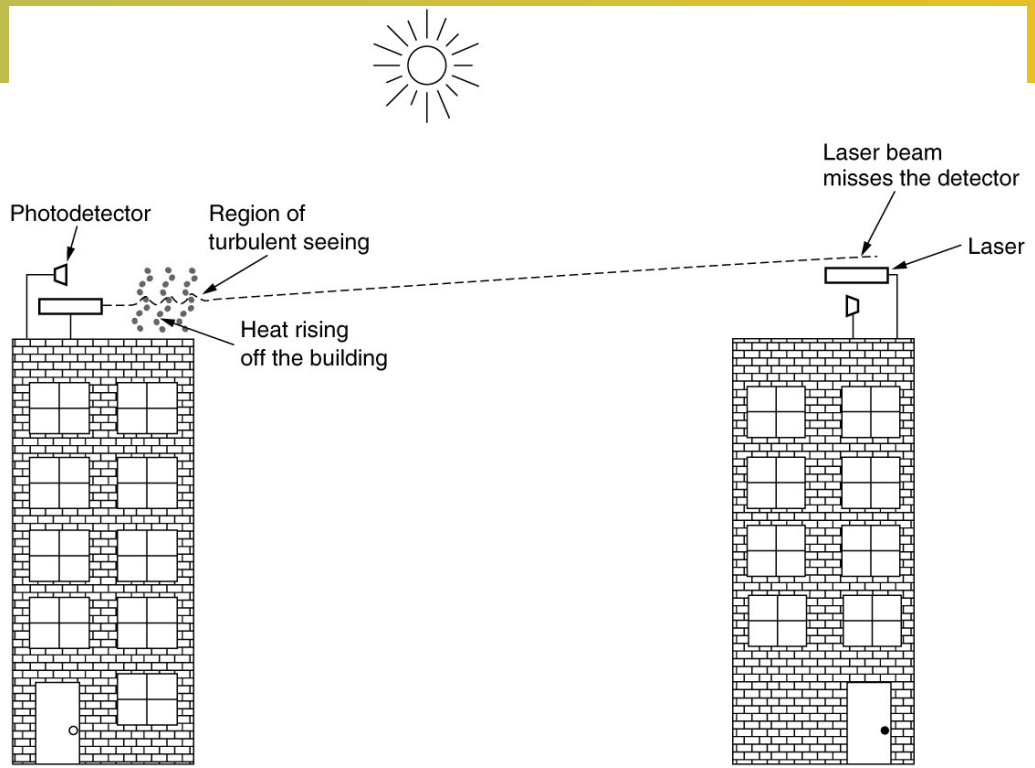


HF band

- In the VLF, LF, and MF bands, radio waves follow the curvature of the earth.
- In the HF band, they bounce off the ionosphere.

https://en.wikipedia.org/wiki/High_frequency

Light Transmission

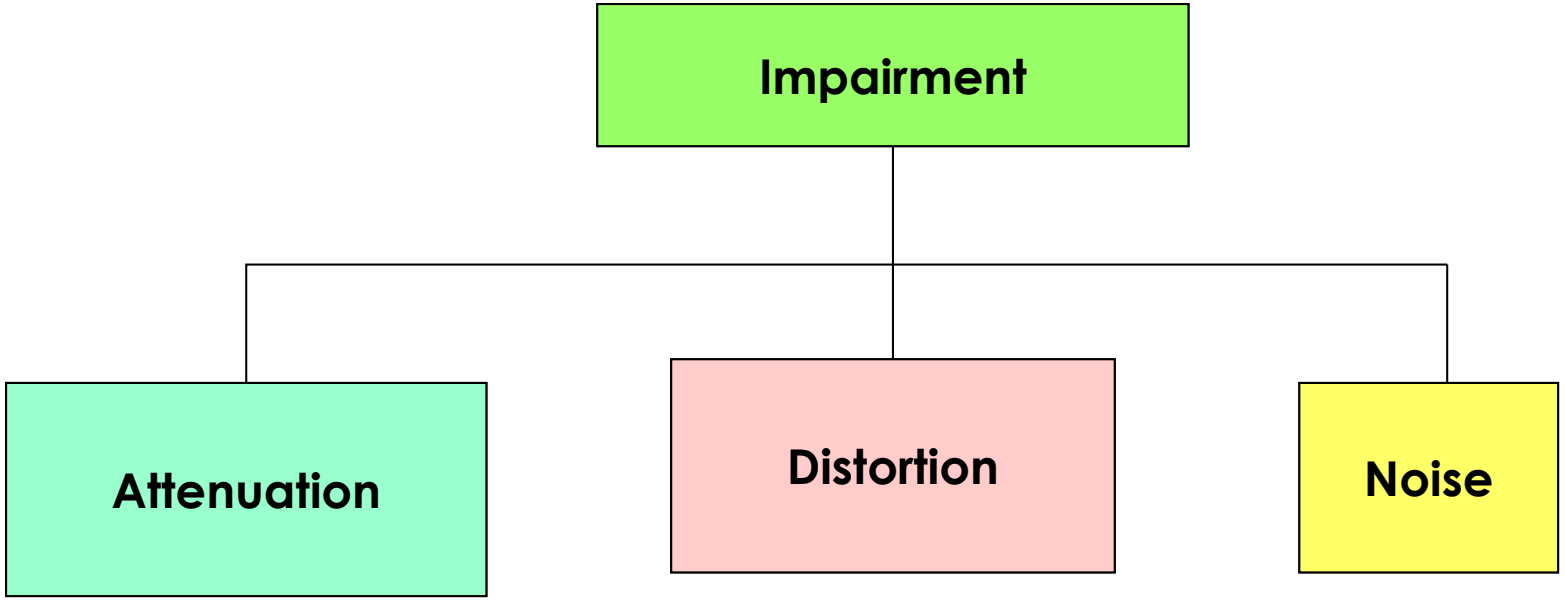


- Convection currents can interfere with laser communication systems.
- A bidirectional system with two lasers is pictured here.

https://en.wikipedia.org/wiki/High_frequency

Transmission Impairment

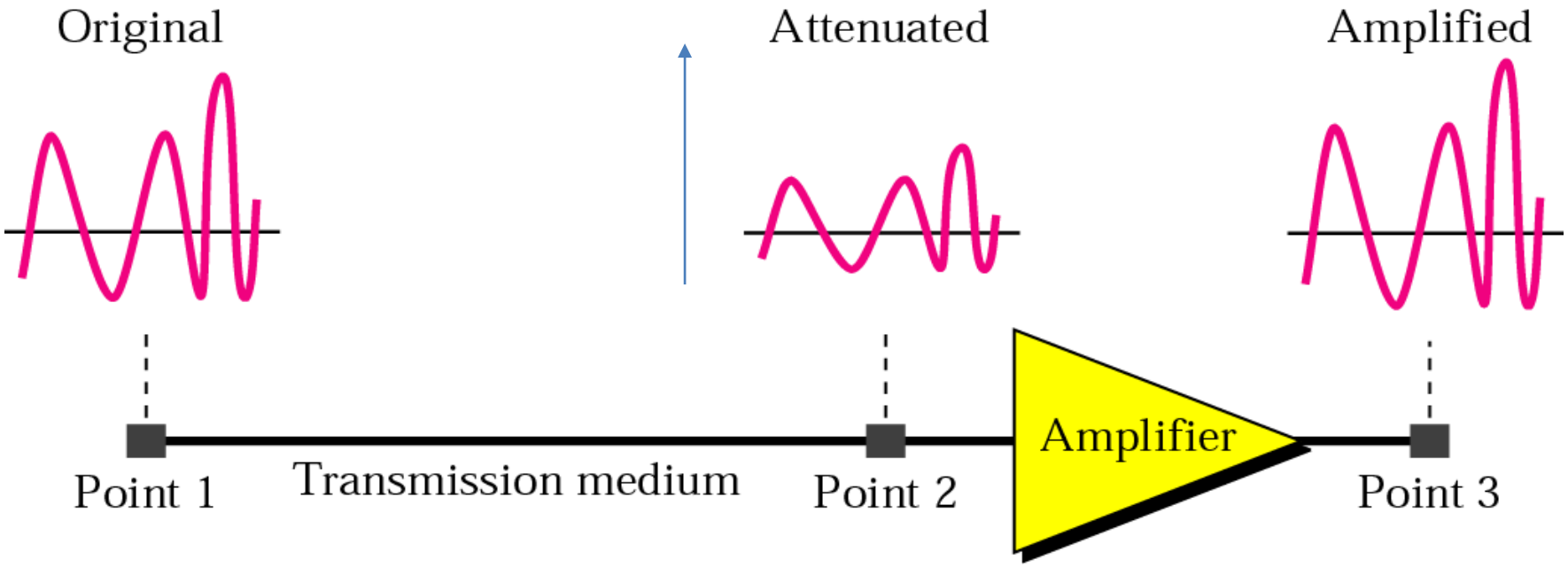
- The imperfections in transmission media cause impairment in the signal



Transmission Impairment

Attenuation

- Attenuation is the loss of signal energy due to the resistance of the transmission media.
- Attenuation limits the length of the transmission media.



Attenuation: Power Ratio

The amount of attenuation can be measured as signal power ratio with respect to the original signal power. The unit of the ratio is decibel.

$$\mathrm{dB} = 10\log_{10}\left(\frac{P_2}{P_1}\right)$$

where

- ✓ P_2 is the signal power at point 2.
- ✓ P_1 is the signal power at point 1.

If dB is negative, we have attenuation ($P_2 < P_1$).
If dB is positive, we have amplification. ($P_2 > P_1$)

Attenuation: Examples

- Imagine a signal travels through a transmission medium and its power reduced to half. This means that $P_2 = \frac{1}{2} P_1$.
  - ✓ In this case, the attenuation can be calculated as

$$\text{dB} = 10 \log_{10}\left(\frac{P_2}{P_1}\right) = 10 \log_{10}\left(\frac{\frac{1}{2}P_1}{P_1}\right) = 10 \log_{10}\left(\frac{1}{2}\right) = -3 \text{ dB}$$

- Imagine a signal travels through a transmission medium and its power is increased by a hundred time. This means that $P_2 = 100\ P_1$.
  - ✓ In this case, the amplification can be calculated as

$$\text{dB} = 10 \log_{10}\left(\frac{P_2}{P_1}\right) = 10 \log_{10}\left(\frac{100P_1}{P_1}\right) = 10 \log_{10}(100) = 20 \text{ dB}$$

Log(2) = 0.3; Log(x/y) = Log(x) - Log(y); Log (x*y) = Log(x)+Log(y)

Attenuation: Examples

- Imagine a signal travels with 1 mW through a transmission medium and its power reduced to 0.2 mW. How many decibels?

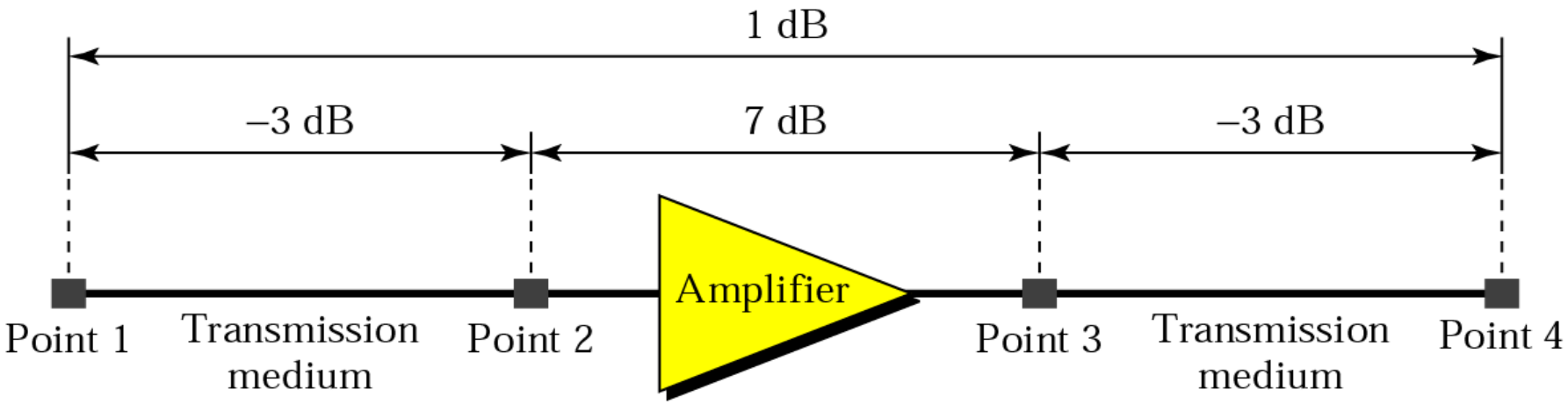
- Sol:

$$\mathrm{dB} = 10\log_{10}\left(\frac{P_2}{P_1}\right) = 10\log_{10}\left(\frac{0.2m}{1m}\right) = 10\log(2*10\text{^}-1)\mathrm{dB}$$

= 10log(2) + 10log(10^-1) = 3-10 = -7 dB

Attenuation: Examples

- What is the signal power ratio in dB between point 4 and point 1?



What is the signal power ratio in dB between point 3 and point 1?
4dB

Attenuation: dBm

- Sometimes the decibel is used to measure signal power in milliwatts.
- In this case, we can represent this absolute measurement (not relative) of power in milliwatts as dBm which can be calculated as follows:

$$\mathrm{dB_m} = 10\log_{10} P_m$$

where P_m is the power in milliwatts.

Example, calculate the power of a signal if its $\mathrm{dB_m}$ is -30.

$$\mathrm{dB_m} = 10\log_{10}\mathrm{Pm} \longrightarrow dB_m = -30 \longrightarrow \log_{10}P_m = -3 \longrightarrow P_m = 10^{-3}\,\mathrm{mW}$$

Note: 0W is not defined

https://www.rapidtables.com/convert/power/dBm_to_Watt.html

Attenuation: Examples

- The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

- Sol:

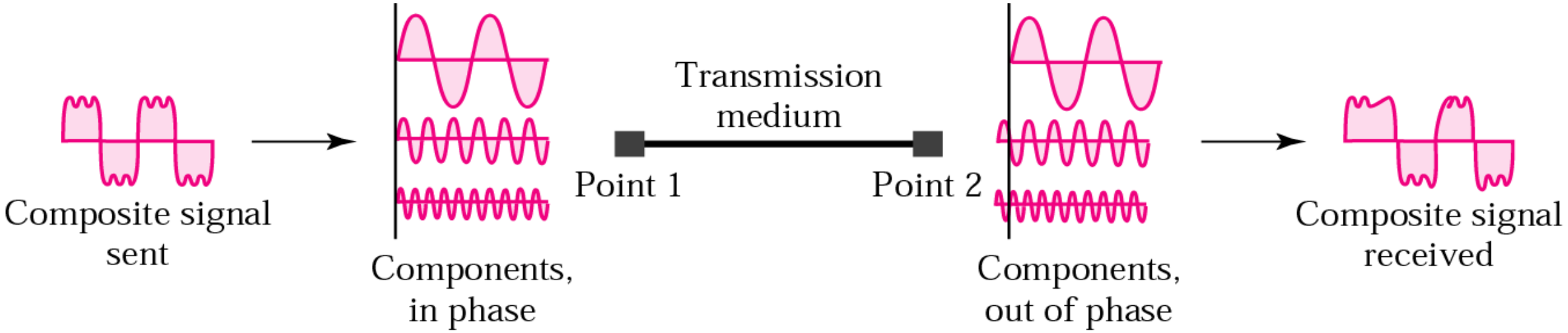
At 5 km, the loss in the cable in decibels should be a 5 block of attenuation system. The loss should be 5 x (-0.3) = -1.5 dB. We then have

$$\text{dB} = 10 \log_{10} (P_2 / P_1) = -1.5 \quad \longrightarrow \quad (P_2 / P_1) = 10^{-0.15} = 0.71$$

$$P_2 = 0.71 P_1 = 0.7 \times 2 \text{ mW} = 1.4 \text{ mW}$$

Distortion

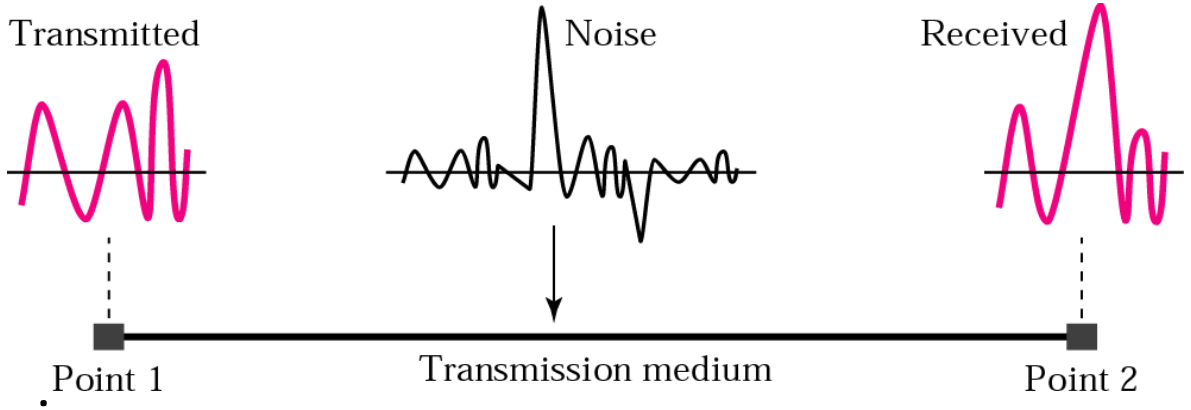
- Distortion occurs in a composite signal, especially digital signals, which is made of different signals with different frequencies. Each signal component has its own propagation speed through a medium ($v = f\lambda$).
- When the signal components of a composite signal arrive at the destination, the characteristics of the recombined signal may change due to various delays of signals.



Transmission Impairment

Noise

- Noise are external signals coming from the environment which can interfere with the transmission of signals carrying data.

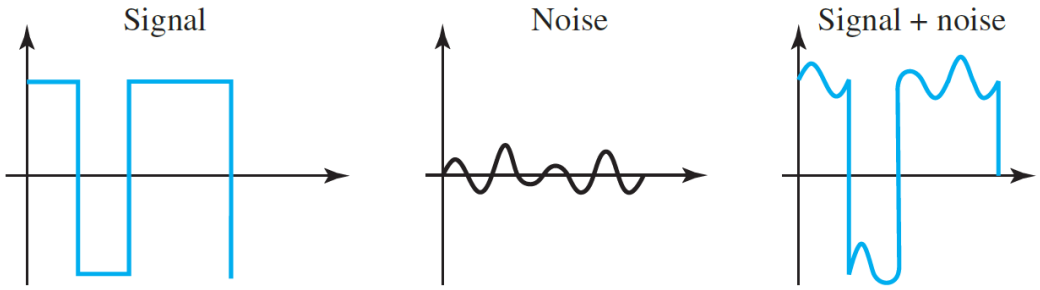


- There are 4 types of noise
  - ✓Thermal noise (from randomly moving electrons)
  - ✓Induced noise (from external magnetic field change)
  - ✓Crosstalk (from magnetic/electrical field change from nearby wires)
  - ✓Impulse noise (from a spike of energy change, e.g., thunder)
- We use Signal-to-Noise ratio (SNR) to measure the effect of noise to the signal (SNR = the power of signal to power of noise)

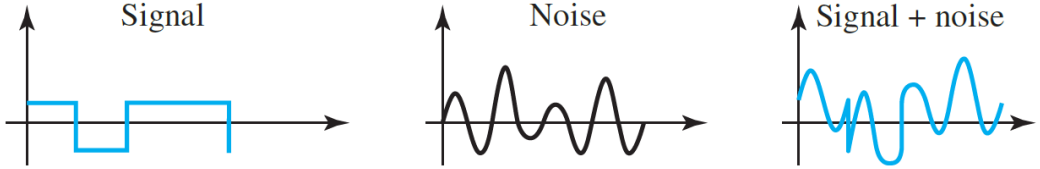
Noise: Signal-to-Noise Ratio (SNR)

- Signal-to-Noise Ratio (SNR) is a measure of the impact of noise to the signal. The signal to noise ratio (SNR) is defined as

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$



a. High SNR



b. Low SNR

Noise: Signal-to-Noise Ratio (SNR)

- SNR is actually the ratio of what is wanted (signal) to what is not wanted (noise).
  - ✓ A high SNR means the signal is less corrupted by noise;
  - ✓ a low SNR means the signal is more corrupted by noise.

- Since SNR is the ratio of two powers, it can also be described in decibel units.

$$SNR_{dB} = 10 \log_{10} SNR$$

Noise: Signal-to-Noise Ratio (SNR) Examples

- **Example 1**: the power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB}?
- Solution:

$$SNR = (10{,}000\ \mu w) / (1\ \mu w) = 10{,}000 \qquad SNR_{dB} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40$$

- **Example 2**: What are the values of SNR and SNR_{dB} for a noiseless channel?
- Solution:

$$SNR = (\text{signal power}) / 0 = \infty \quad \longrightarrow \quad SNR_{dB} = 10 \log_{10} \infty = \infty$$

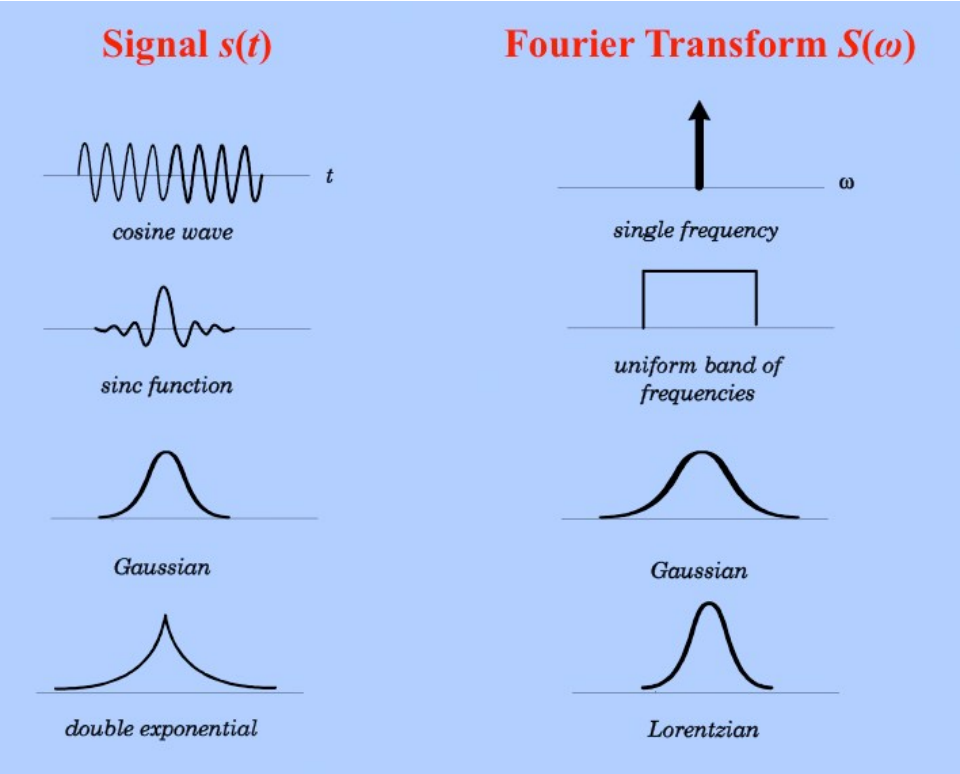
From Waveforms to Bits

- The theoretical basis for data communication
  - Fourier analysis
  - Bandwidth-limited signals

- The maximum data rate of a channel

Frequency Domain Transformation

- Sometimes, we don't want to draw a lot to describe signals. So, we change our representation of waveforms to representation of waveforms in terms of sine waves where a sine wave of a particular frequency staring at original can be represented by a vertical line in **frequency domain** plot.



Time domain Frequency domain

Fourier Analysis

- We model the behavior of variation of voltage or current with mathematical functions
- Fourier series is used

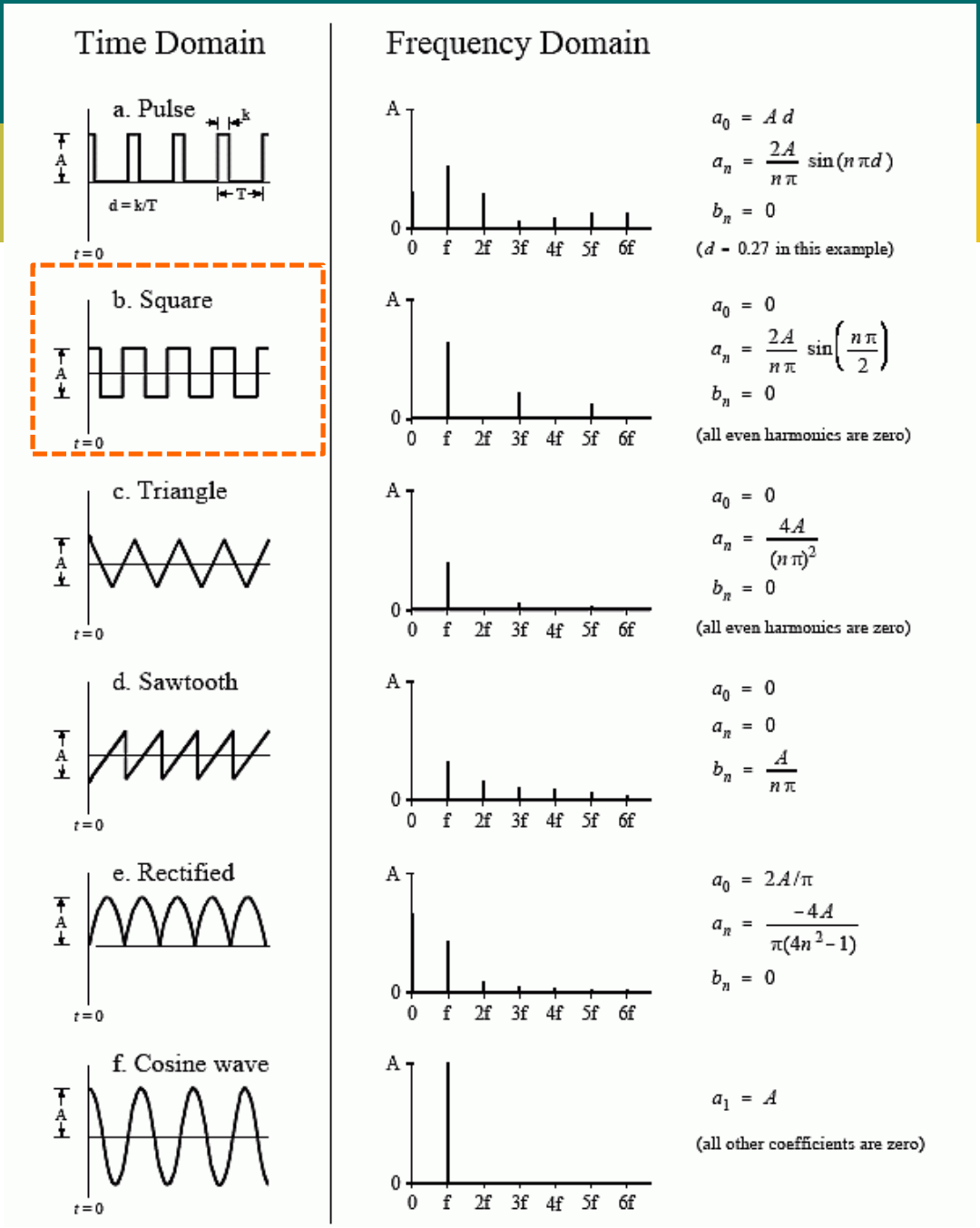
$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

- Function reconstructed with

$$a_n = \frac{2}{T}\int_0^T g(t)\sin(2\pi nft)\,dt \qquad b_n = \frac{2}{T}\int_0^T g(t)\cos(2\pi nft)\,dt \qquad c = \frac{2}{T}\int_0^T g(t)\,dt$$

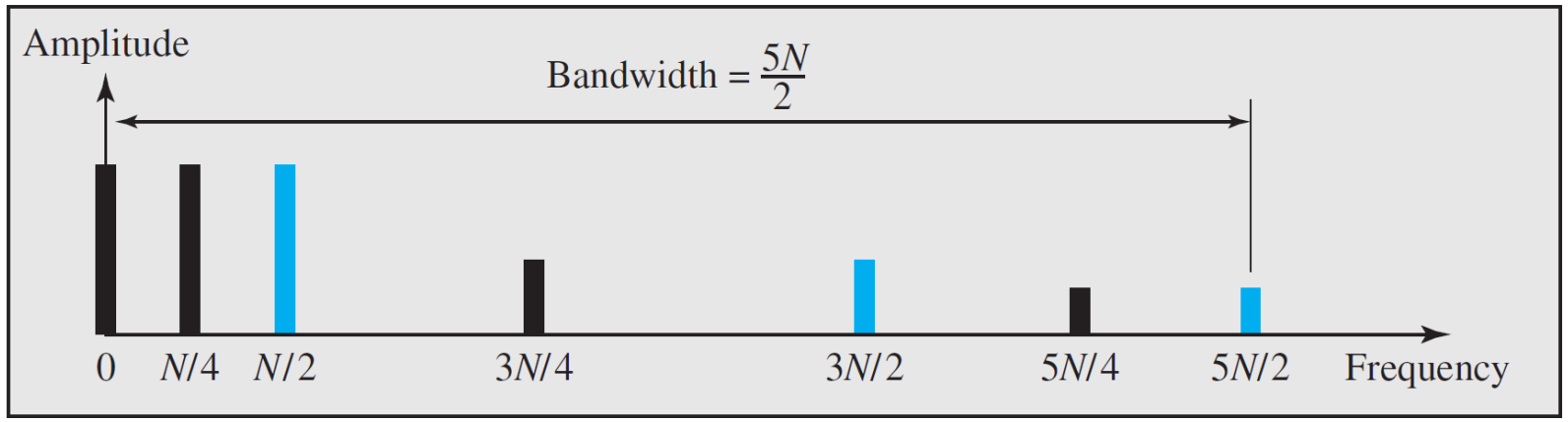
From Waveforms to Bits

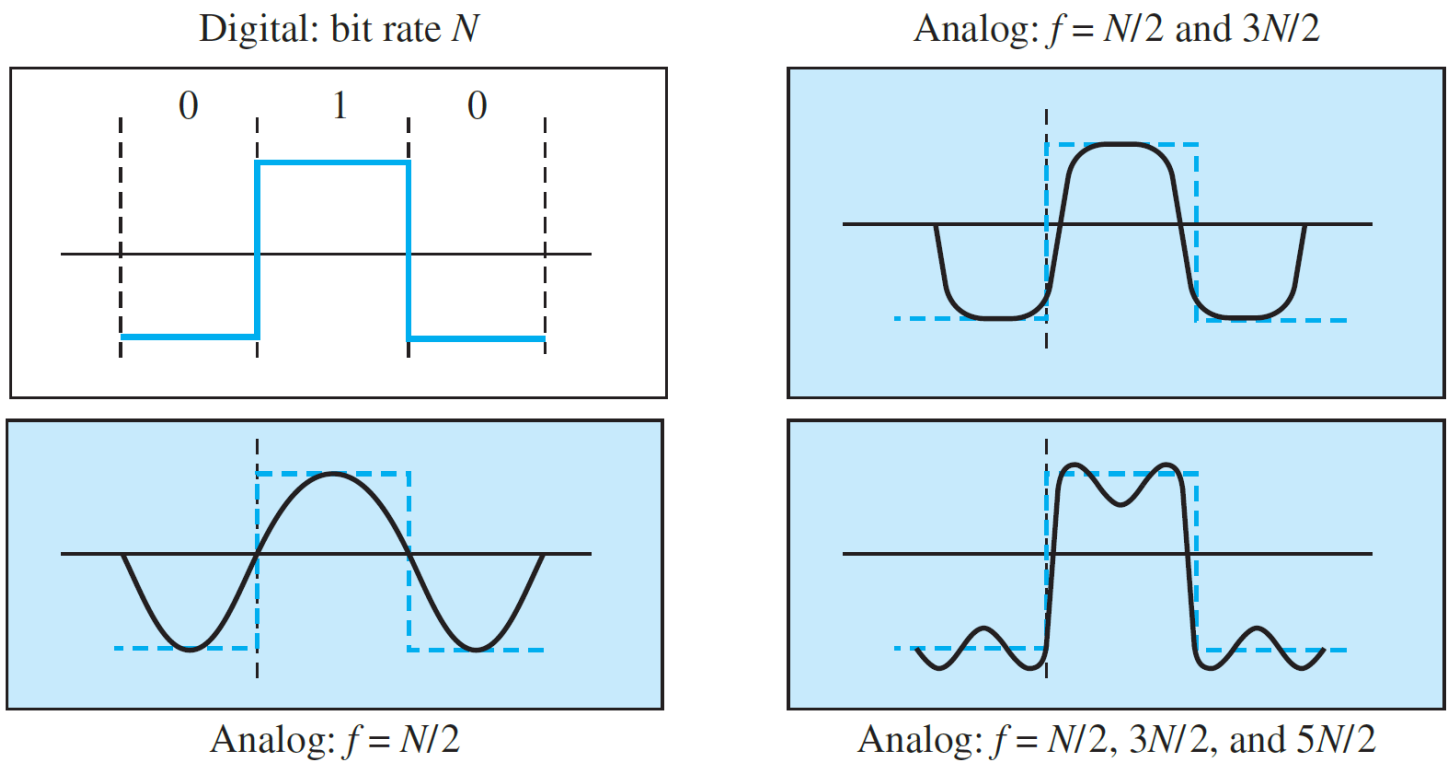
Fourier Transform



From Waveforms to Bits

Digital Signals





From Waveforms to Bits

Noise

Square wave => data

Noise

Noise+Data

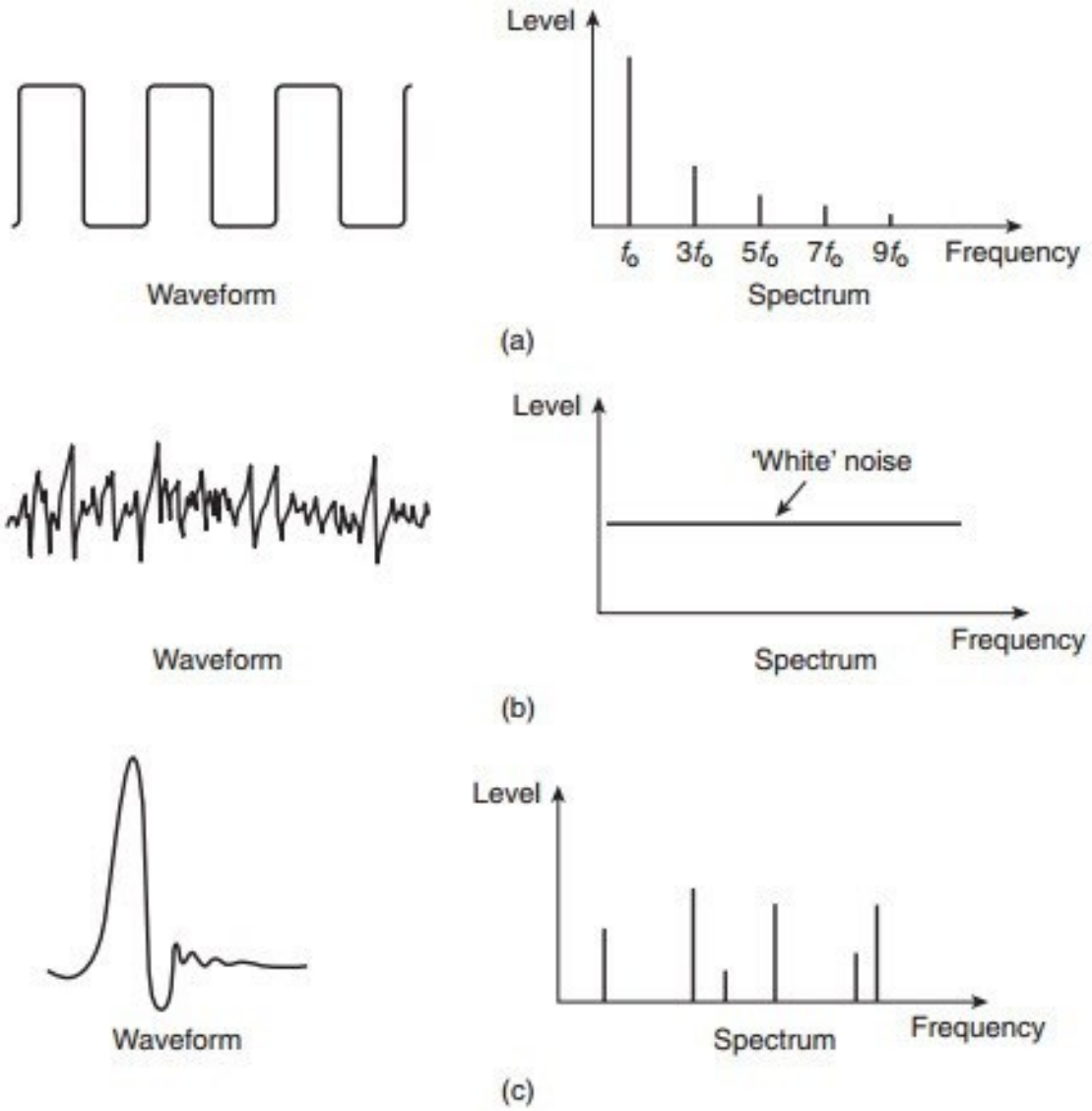
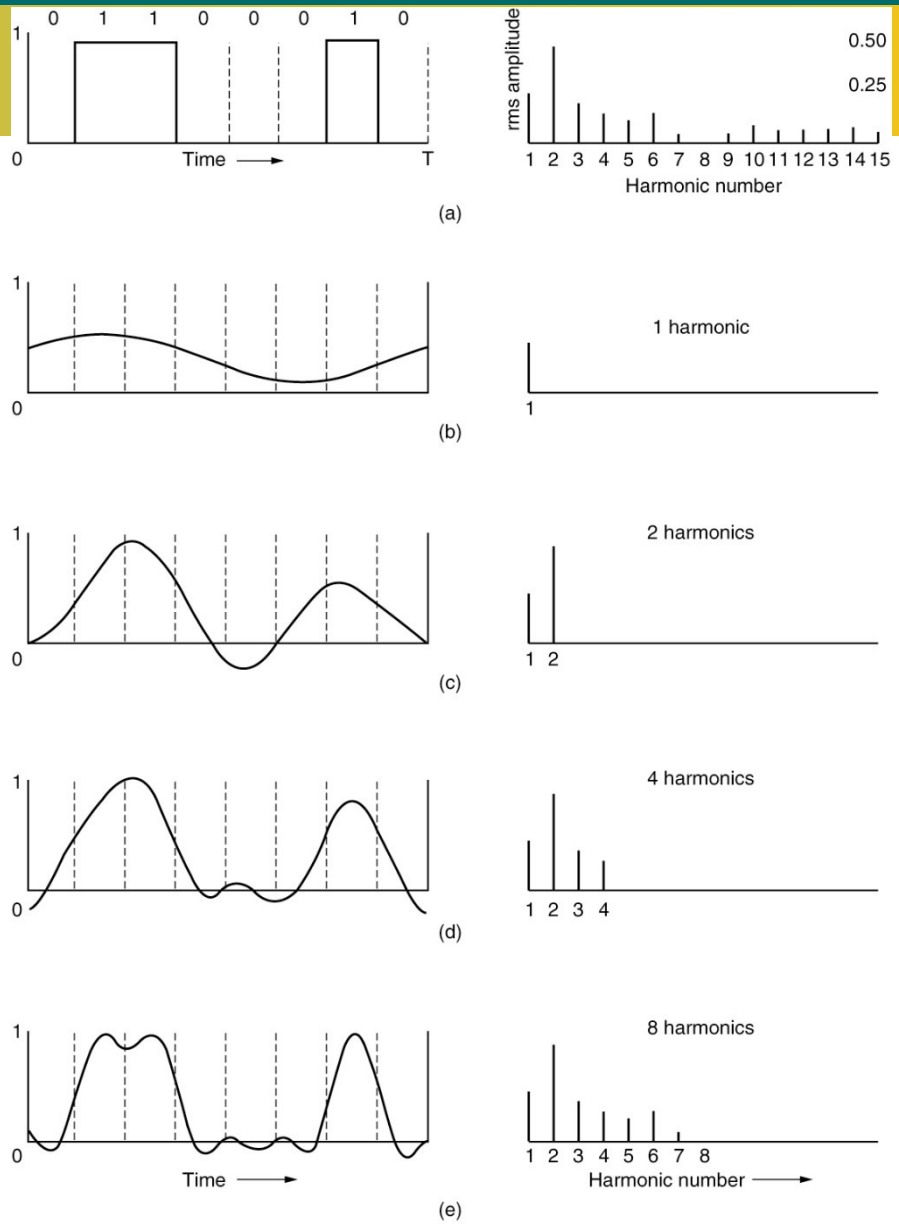


Figure 1.3: (a) A periodic signal repeats after a fixed time and has a simple spectrum consisting of fundamental plus harmonics. (b) An aperiodic signal such as noise does not repeat and has a continuous spectrum. (c) A transient contains an anharmonic spectrum.

Bandwidth-Limited Signals

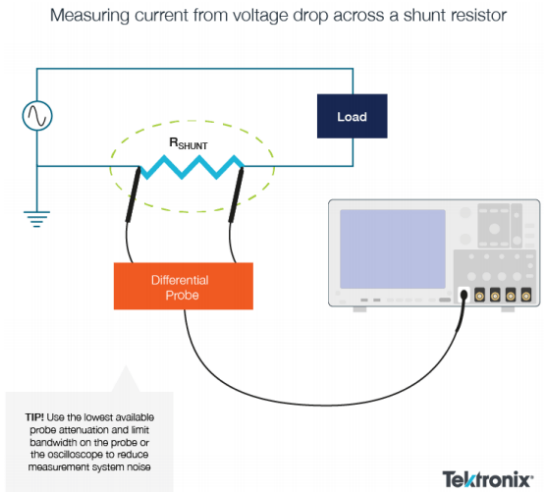
- A binary signal and its root-mean-square Fourier amplitudes. This is followed by successive approximations to the original signal.
- The relation between data rate and harmonics for our example.

Bps	T (msec)	First harmonic (Hz)	# Harmonics sent
300	26.67	37.5	80
600	13.33	75	40
1200	6.67	150	20
2400	3.33	300	10
4800	1.67	600	5
9600	0.83	1200	2
19200	0.42	2400	1
38400	0.21	4800	0

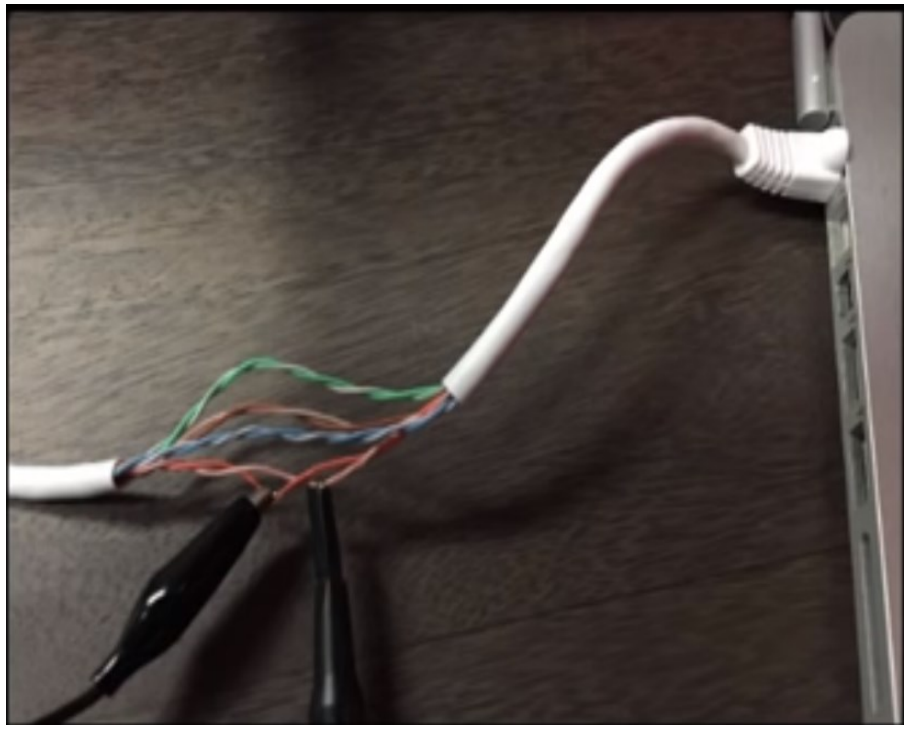


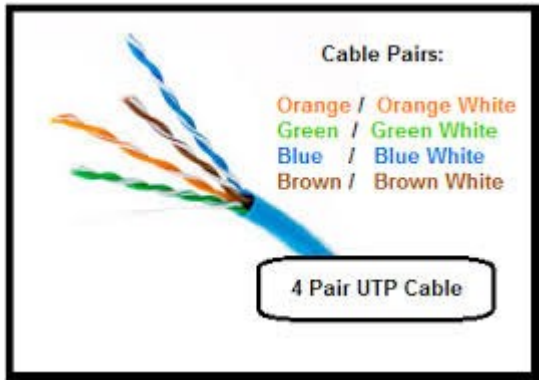
Signaling Measurement

- We can measure signals (usually electrical signals) using a device called Oscilloscope.
- In this case, the transmission of data signal over Ethernet cable



- Setup:



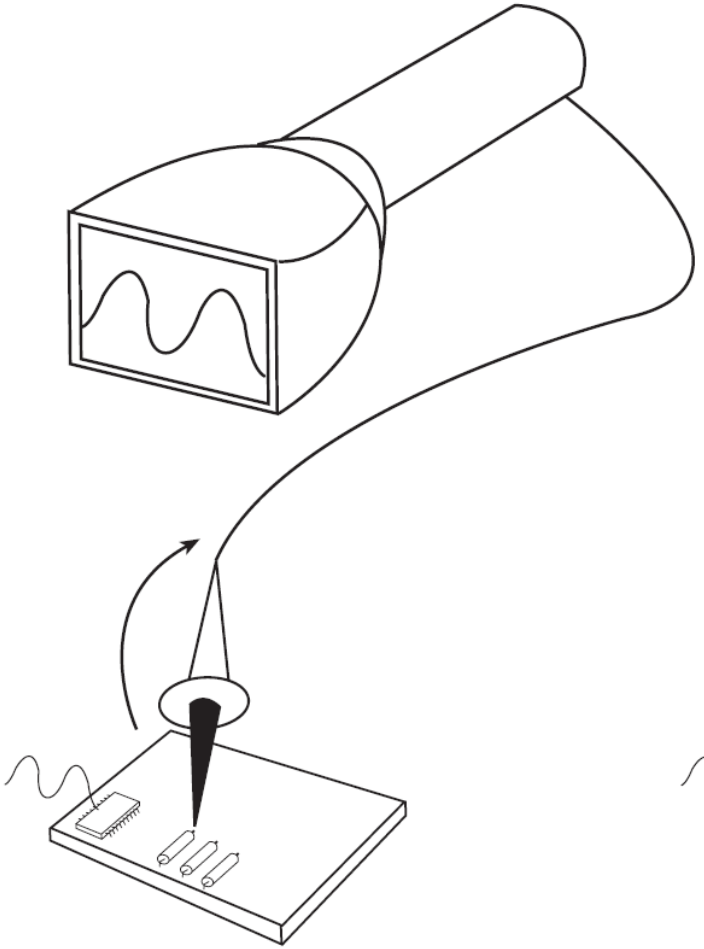


Oscilloscope

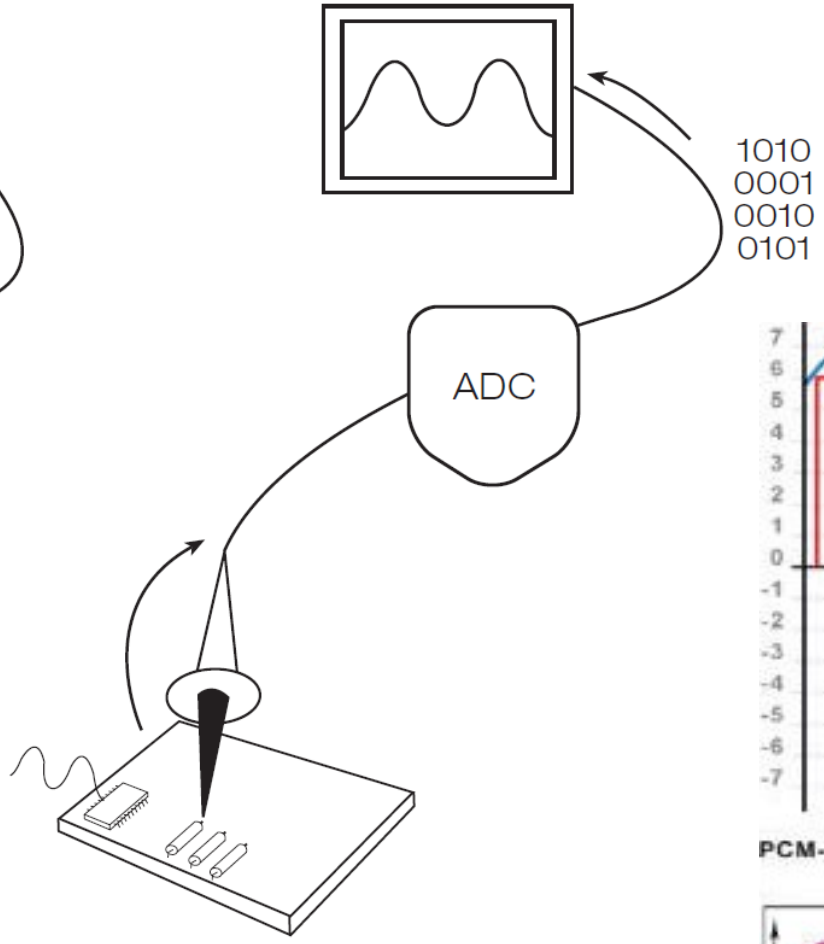
- previously called an oscillograph, and informally known as a scope or o-scope, is a type of electronic test instrument that graphically displays varying signal voltages, usually as a two-dimensional plot of one or more signals as a function of time.
- An oscilloscope displays a graph showing the voltage snapshots of an electrical signal over time, with voltage as the Y-axis and time as the X-axis.
- A simple graph can tell us many things about a signal such as
  - ✓ The time and voltage values of a signal
  - ✓ The frequency of an oscillating signal
  - ✓ The frequency with which a particular portion of the signal is occurring relative to other portions
  - ✓ Whether or not a malfunctioning component is distorting the signal
  - ✓ How much of a signal is direct current (DC) or alternating current (AC)
  - ✓ How much of the signal is noise and whether the noise is changing with time

Signaling Measurement

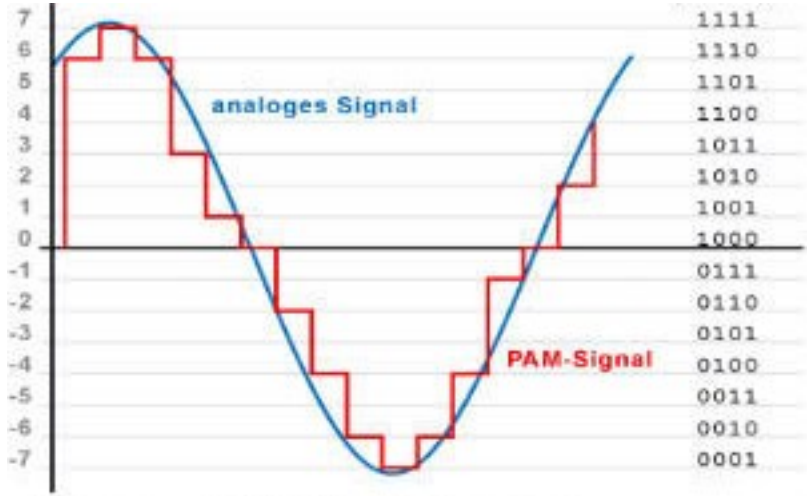
Oscilloscope

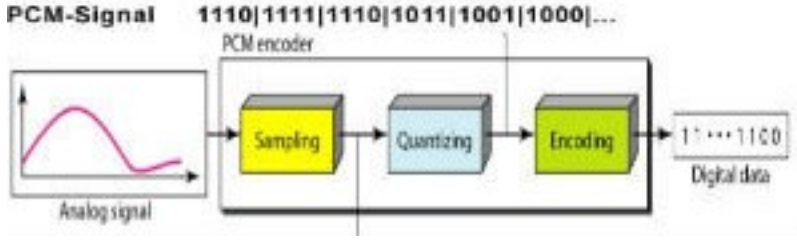


Analog Oscilloscopes
Trace Signals



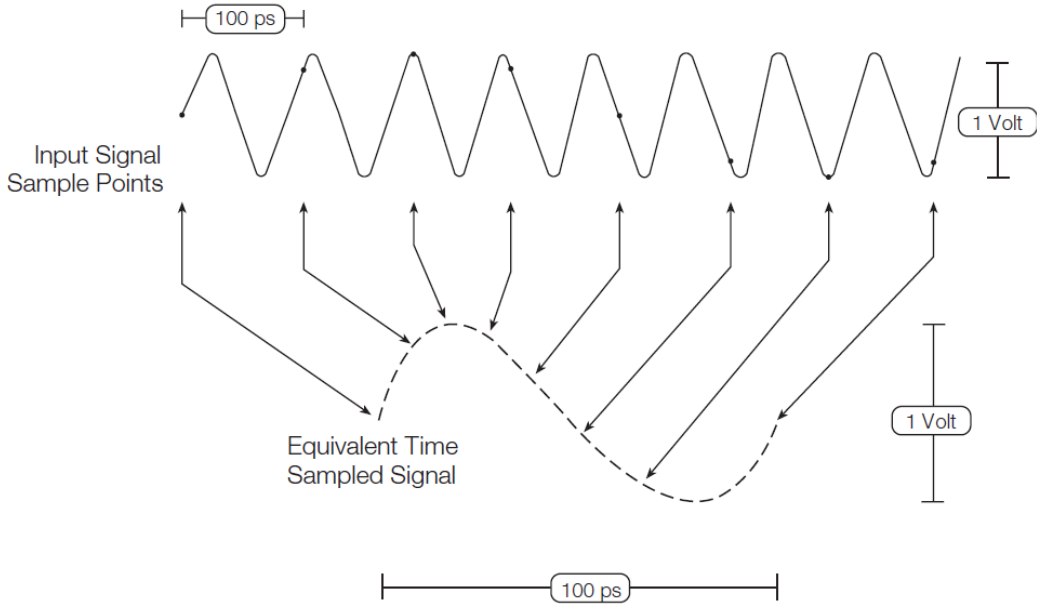
Digital Oscilloscopes Samples
Signals and Construct Displays





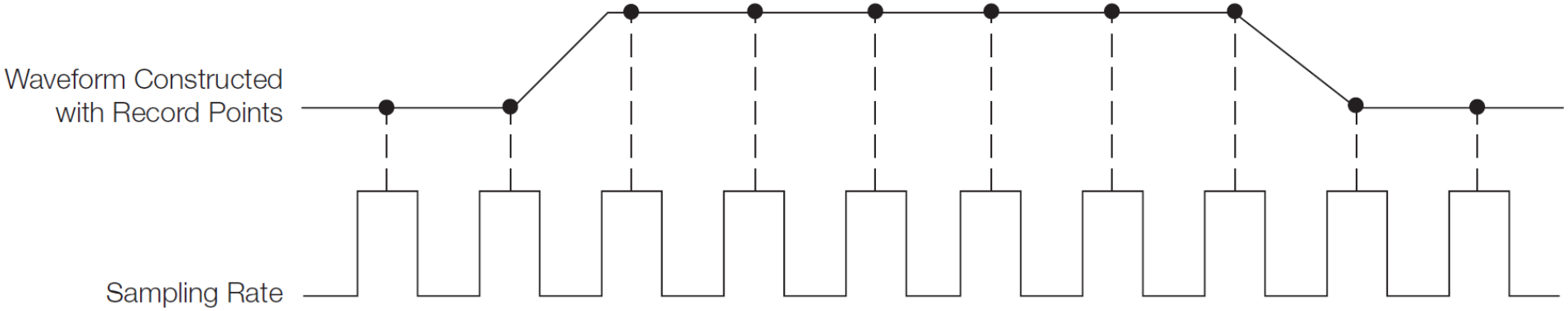
Sampling (1/2)

- It is impossible to measure the levels of signal all the time. We have to take snapshots of signals at various time.
- Oscilloscope shows graph of signal voltages by taking snapshots of the original signal in rapid repeated fashion. This process is called sampling.
- Sampled points are connected by interpolation to produce a continuous waveform.



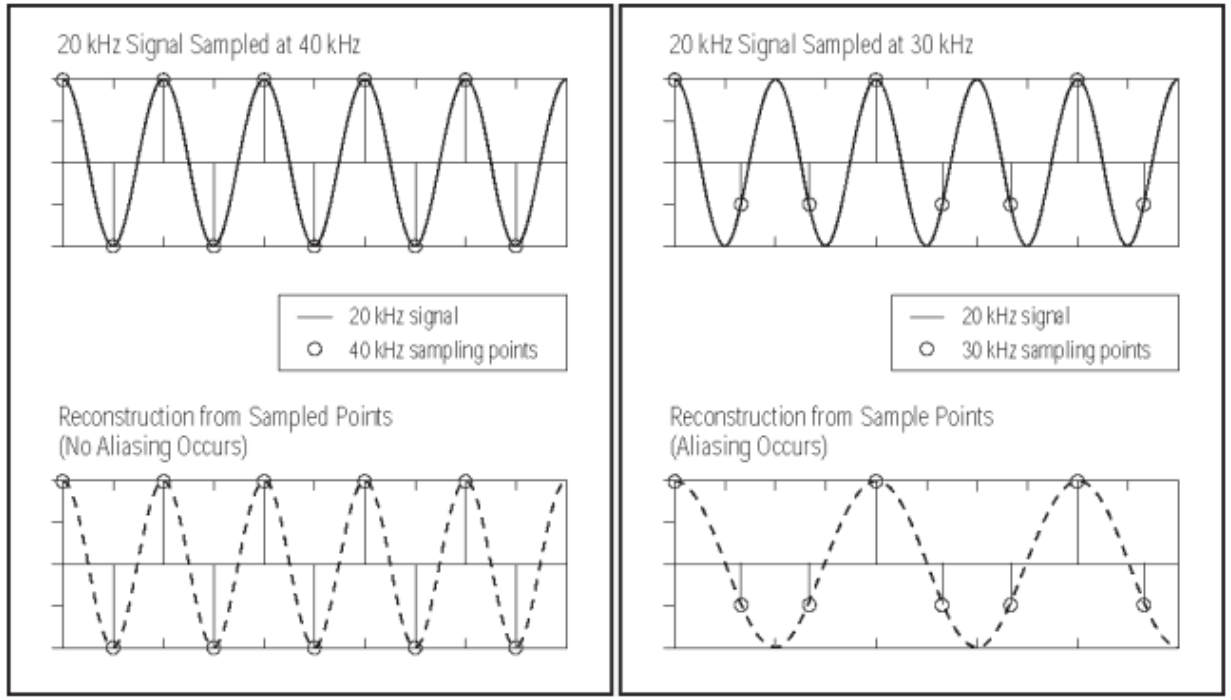
Sampling (2/2)

- Sampling is the process of taking multiple measurements of the signal. It can also be described as a process of converting a portion of an input signal into a number of discrete electrical values for the purpose of storage, processing and/or display.
- The magnitude of each sampled point is equal to the amplitude of the input signal at the instant in time in which the signal is sampled.
- Sampling is like taking snapshots of a signal. Each snapshot corresponds to a specific point in time on the waveform. These snapshots can then be arranged in the appropriate order in time so as to reconstruct the input signal.
- In order to accurately measure the signal waveform, we must sample the signal at adequate frequency.



Sampling Rate

- Sample rate – specified in samples per second (S/s) – refers to how frequently a digital oscilloscope takes a snapshot or sample of the signal, analogous to the frames on a movie camera.
- The faster an oscilloscope samples (i.e., the higher the sample rate), the greater the resolution and detail of the displayed waveform and the less likely that critical information or events will be lost
- **The minimum sample rate** may also be important if you need to look at slowly changing signals over longer periods of time.



Sampling Rate Requirements

- In order to accurately reconstruct a signal and avoid aliasing, Nyquist theorem says that the signal must be sampled **at least twice** as fast as its **highest frequency** component.
- Note: This theorem, however, assumes an infinite record length and a continuous signal. Since no oscilloscope offers infinite record length and, by definition, glitches are not continuous, sampling at only twice the rate of highest frequency component is usually insufficient.
  - ✓ Record length, expressed as the number of points that comprise a complete waveform record, determines the amount of data that can be captured with each channel
  - ✓ Since an oscilloscope can store only a limited number of samples, the waveform duration (time) will be inversely proportional to the oscilloscope's sample rate.

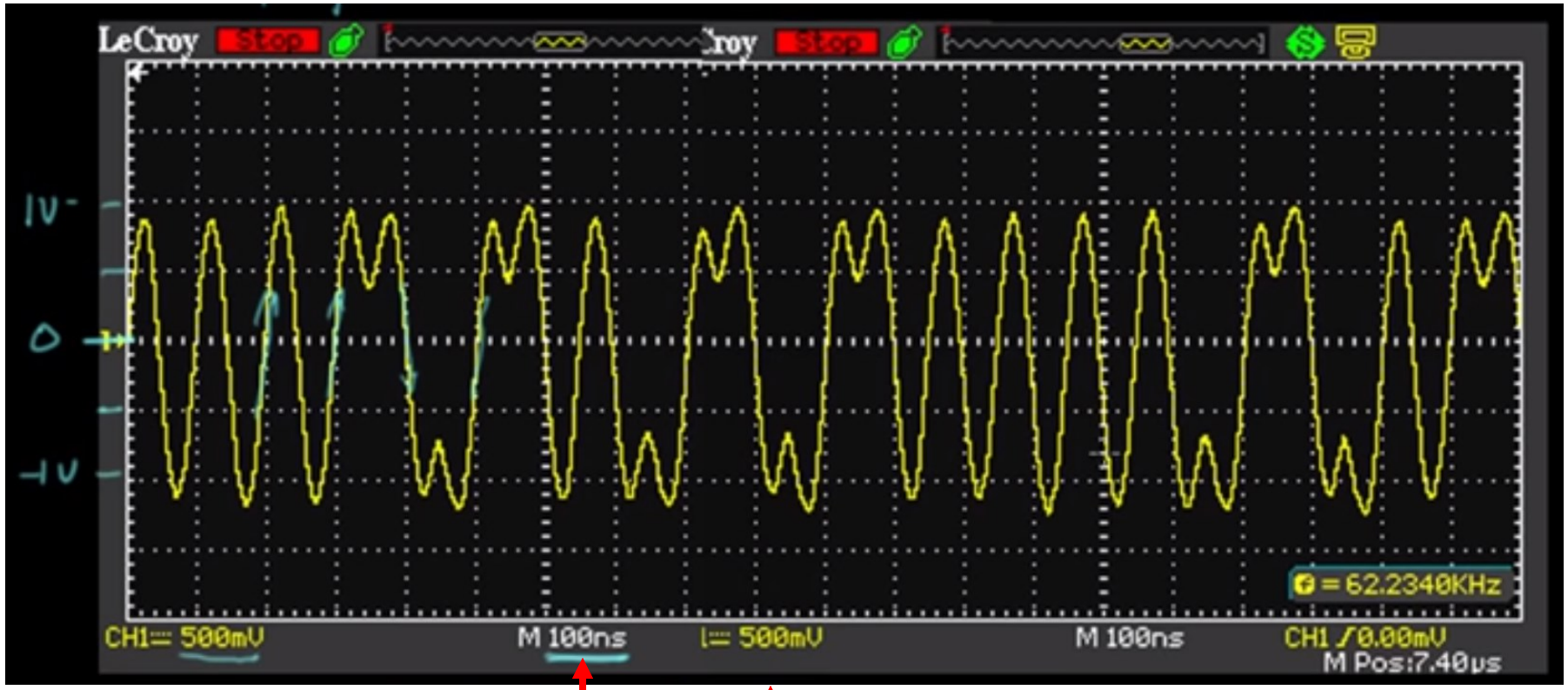
$$\text{Time Interval} = \frac{\text{Record Length}}{\text{Sample Rate}}$$

Voice data: 0-4KHz
Sampling Rate 2*4K=8KHz

Signaling Measurement

Example

- Example: Computer transmits data 01001011 out to a destination
- The signal detected is shown below

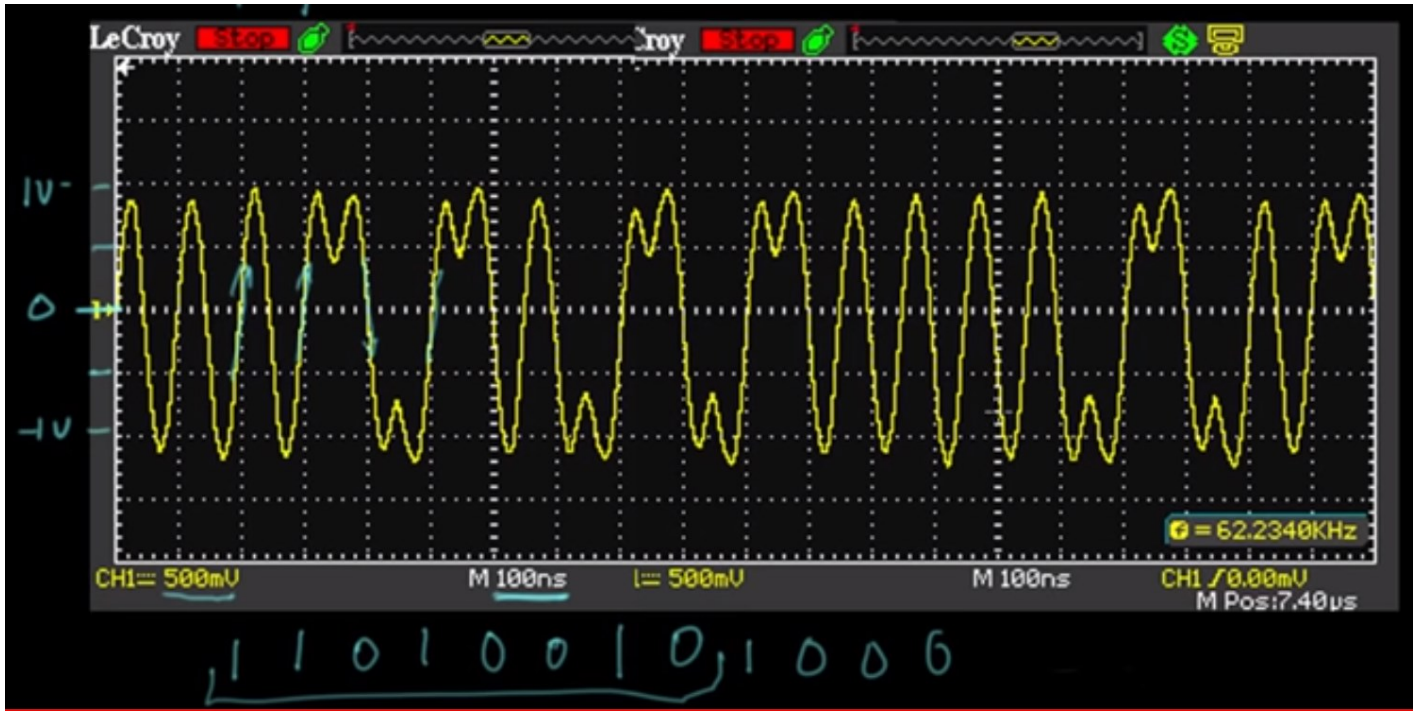


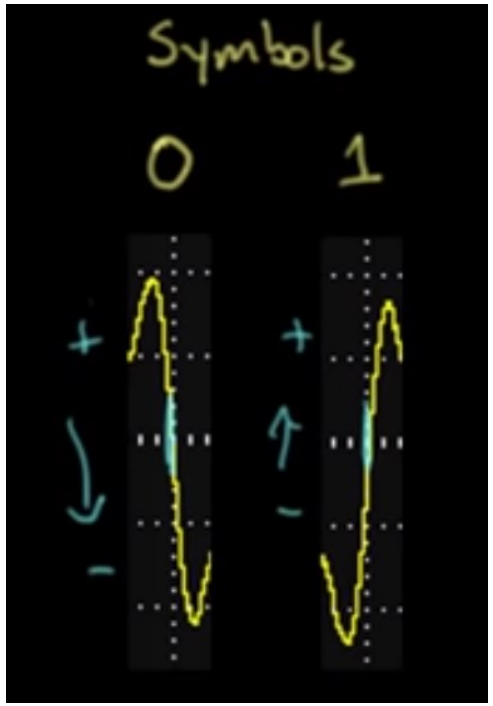
1 horizontal step = 100 ns

1 vertical step = 500 mV

Example: Data Reading

- Using Manchester encoding (semantics), we can read the data as follows:





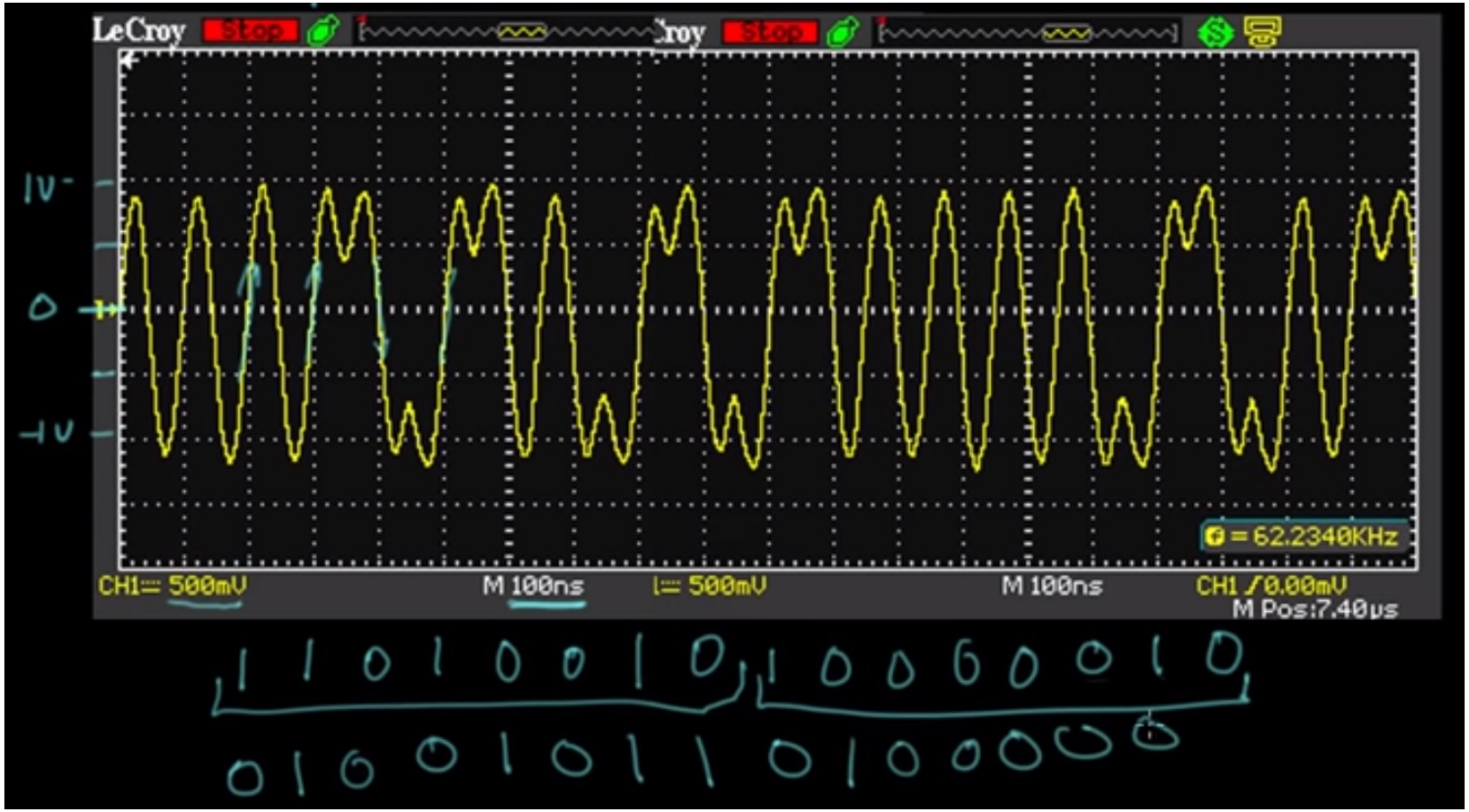
1 horizontal step = 100 ns

One symbol must be read once every cycle (equal period).

Each symbol or part of symbol must not be read twice.

Example: Data Interpretation

- The higher value of time shows the data/signals which have been sent out earlier.
- Therefore, the stored data must be read backwards since the values to the right arrive first.



MSB LSB

11010010

Data Rate vs Signal Rate

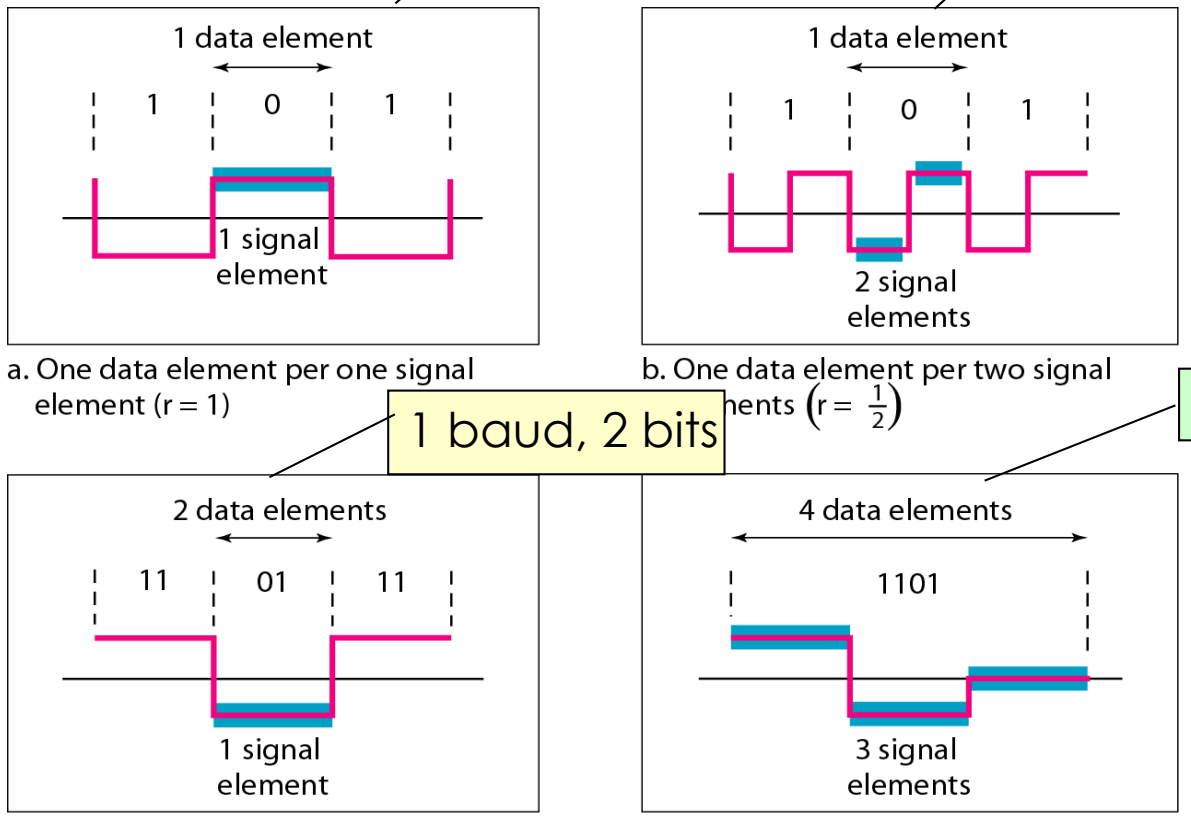
- Data rate
  - It is the number of data elements (bits) sent in 1 second.
  - It is also called bit rate.

- Signal rate
  - It is the number of signal elements (symbols) sent in 1 second.
  - It is also called baud rate or baud or pulse rate or modulation rate.

Data Rate vs Signal Rate

1 baud, 1 bit

1 baud, half bit



a. One data element per one signal element (r = 1)

b. One data element per two signal elements $\left(r = \frac{1}{2}\right)$

1 baud, 2 bits

3 bauds, 4 bits

c. Two data elements per one signal element (r = 2)

d. Four data elements per three signal elements $\left(r = \frac{4}{3}\right)$

Data Rate vs Signal Rate

- According to the relationship between signal rate and data rate below, our goal is to increase the data rate while decreasing the signal rate in order to minimize the energy usage and actual signal transmission time.

$$S = c \times N \times \frac{1}{r}$$

- ✓ S = number of signal element (signal rate)
- ✓ c = case factor (0 == best; 1 == worst)
- ✓ N = data rate = bit rate
- ✓ r = bits per baud or bits per signal

Data Rate vs Signal Rate

Example

- A signal is carrying data in which one data element is encoded as one signal element (r = 1). If the bit rate is 100 kbps, what is the average value of the baud rate if c is 1/2?
- Sol:

$$S = c \times N \times \frac{1}{r} = \frac{1}{2} \times 100{,}000 \times \frac{1}{1} = 50{,}000 = 50 \text{ kbaud}$$

Data Rate vs Signal Rate

Bandwidth Requirements

- Although the actual bandwidth of a digital signal is infinite, the effective bandwidth is finite.

$$\mathrm{B_{min}} = S = c \times N \times \frac{1}{r}$$

$$\mathrm{N_{max}} = r \times B \times \frac{1}{c}$$

- ✓ B_{min} = minimum bandwidth = baud rate
- ✓ N_{max} = maximum data rate
- ✓ c = case factor
- ✓ r = bits per baud or bits per symbol

Data Rate vs Signal Rate

Bandwidth Requirements: Example

- The maximum data rate of a channel is $N_{max} = 2 \times B \times \log_2 L$ (defined by the Nyquist formula). Does this agree with the previous formula for N_{max}?

- Solution:
  - A signal with L levels actually can carry $\log_2 L$ bits per level ($r = \log_2 L$). If each level corresponds to one signal element and we assume the average case ($c = 1/2$), then we have

$$N_{\max} = \frac{1}{c} \times B \times r = 2 \times B \times \log_2 L$$

$$\text{bit rate} = 2 \times \text{baud rate} \times \log_2 L$$

Bandwidth Requirements: Examples

- Two signals (A and B) have two and four data levels with a pulse duration of 1 ms. The pulse rates and bit rates of the signals are as follows

- Signal A : **2** data levels

$$\text{pulse rate} = B = \frac{1}{1\ \text{ms}} = \frac{1}{10^{-3}} = 1{,}000\ \text{baud}$$

$$\text{bit rate} = N = 2 \times B \times \log_2 L = 2 \times 1{,}000 \times \log_2 2 = 2{,}000\ \text{bps}$$

- Signal B : **4** data levels

$$\text{pulse rate} = B = \frac{1}{1\ \text{ms}} = \frac{1}{10^{-3}} = 1{,}000\ \text{baud}$$

$$\text{bit rate} = N = 2 \times B \times \log_2 L = 2 \times 1{,}000 \times \log_2 4 = 4{,}000\ \text{bps}$$

The Maximum Data Rate of a Channel

Data Rate Limit

- How fast can we send data?
- Data rate depends on 3 factors:
  - Bandwidth available of the channel
  - Level of Signals we use
  - Quality of medium or channel (level of noise in the channel)
- In order to calculate data rate appropriate for each channel, we consider
  - **Nyquist's** theorem for noiseless channel

    Maximum data rate (*bits/sec*) = $2B\log_2 V$ bits/sec → V = L

  - **Shannon's** formula for capacity of a noisy channel

    Maximum data rate {*bits/sec* }= $B\log_2 (1 + S/N)$

Nyquist Bit Rate

- Nyquist bit rate is the theoretical maximum bit rate for the channel.
- Assume there is **no noise** (=noiseless) in the channel.

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

- ✓ Bandwidth is the bandwidth of the channel
- ✓ **L** is the number of **signal levels** used to represent data
- ✓ BitRate is the bit rate in bits per second

- ✓ Note: the more levels we use, the more difficult it is for the receiver to distinguish the signal voltage of each level.

Nyquist Bit Rate

Examples

Example 3.34

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Example 3.35

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12{,}000 \text{ bps}$$

Nyquist Bit Rate

Examples

Example 3.36

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as shown:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L \longrightarrow \log_2 L = 6.625 \longrightarrow L = 2^{6.625} = 98.7 \text{ levels}$$

Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate. If we have 128 levels, the bit rate is 280 kbps. If we have 64 levels, the bit rate is 240 kbps.

Shannon Bit Rate

- In reality, the channel is always **noisy**. We must take into account, the level of noise in the channel.

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

- ✓ Bandwidth is the bandwidth of the channel.
- ✓ SNR is the signal level to noise level ratio (usually in dB)
- ✓ Capacity is the capacity of the channel in bits per second.

Shannon Bit Rate

Examples

Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2(1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Example 3.38

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2(1 + 3162) = 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Shannon Bit Rate

Examples

Example 3.39

The signal-to-noise ratio is often given in decibels. Assume that $SNR_{dB} = 36$ and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$SNR_{dB} = 10 \log_{10} SNR \longrightarrow SNR = 10^{SNR_{dB}/10} \longrightarrow SNR = 10^{3.6} = 3981$$

$$C = B \log_2(1 + SNR) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

Example 3.40

When the SNR is very high, we can assume that SNR + 1 is almost the same as SNR. In these cases, the theoretical channel capacity can be simplified to $C = B \times SNR_{dB}$. For example, we can calculate the theoretical capacity of the previous example as

let's proof this

$$C = 2 \text{ MHz} \times (36 / 3) = 24 \text{ Mbps}$$

Shannon Bit Rate

Examples

- From the Shannon Bit Rate formula

$$C = B \times \log_2(1 + SNR)$$

- for large SNR, SNR+1 ~ SNR
- We have

$$C = B \times \log_2(SNR)$$

$$C = B \times \frac{\log_{10} SNR}{\log_{10} 2}$$

- Since

$$SNR_{db} = 10 \log_{10} SNR$$

- We have

$$C = B \times \frac{\log_{10} SNR}{\log_{10} 2} = B \times \frac{10}{10} \times \frac{\log_{10} SNR}{\log_{10} 2} = B \times \frac{10 \log_{10} SNR}{10 \log_{10} 2} = B \times \frac{SNR_{db}}{3}$$